

wasted clock cycles as each core has a much better chance of being kept busy due to the fact that each core can be fed with two threads at a time, instead of a single thread. Nehalem's cores are much faster than those in a Pentium 4, partly due to their being derived from the much faster Core architecture and also due to the much shorter execution pipeline; this means that HT on an i7 should be much more efficient than HT was on the Pentium 4 CPUs.

Intel's Penryn CPUs added support for SSE 4 and Nehalem brings with it support for SSE 4.2. The main additions with the new SSE instructions are improved XML processing support, voice-recognition, error-detection code calculation, and DNA sequencing. What we're seeing here is greater emphasis on achieving more performance per watt. While Nehalem CPUs have a higher TDP (Thermal Dissipation Point) than Penryn at 130 watts versus 95 watts, Intel claims that the performance hike is more than proportionate to the increased power requirement, or at worst, equal to it. In fact the Nehalem is designed on the Atom processor ethos: that is to maintain a 2:1 ratio for increments in performance and power consumption figures at all times.

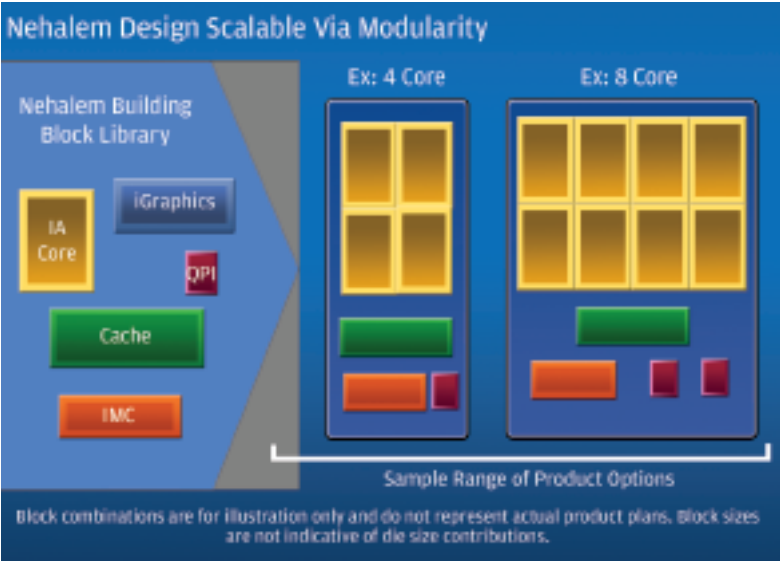
Nehalem is a native quad-core with four cores sitting on a single die. Unlike previous quad-cores, wherein two cores shared an L2 cache, i7 offers each core discrete L1 and L2 caches, with an L3 cache in common. The Core i7 is made from a budget of 731 million transistors. In comparison, Yorkfield needed 820 million transistors. The die size of Nehalem is larger though; increasing from 214 mm² to 263 mm². That's fewer transistors and reduced density, which should bring thermal benefits. Nehalem has the same L1 cache as previous Yorkfield CPUs: a 32 KB instruction cache and 32 KB data cache per core, but the L2 cache is significantly smaller. Each core in a quad-core Nehalem CPU gets only 256 KB (1 MB in total), while Yorkfield CPUs had as much as 3 MB of L2 cache per core. There is a single 8 MB of L3 cache to which all cores have access.

Copycat?

Now why did Intel make a high-end CPU with a fraction of the L2 cache? And why introduce an L3 cache which we know is slower than any L2 cache? The answer is one of the most radical architectural differences between Core i7 and Core 2—the inclusion of an on-die memory controller or integrated memory controller (IMC). Gone is the

Nehalem vs. Penryn: cache and memory		
	Nehalem	Penryn (Yorkfield)
L1 Size / L1 Latency	64 KB / 4 cycles	64 KB / 3 cycles
L2 Size / L2 Latency	256 KB / 11 cycles	12 MB / 15 cycles
L3 Size / L3 Latency	8 MB / 39 cycles	NA
Main Memory Latency *	107 cycles (33.4 nano sec)	160 cycles / 50.3 nano sec

* Using DDR3 1600 MHz (CAS Latency 7)



memory controller on the Northbridge. This is where AMD fans will scream “copycat”. That’s not all. Gone is the ancient, bi-directional Front-Side Bus (FSB) in favour of what Intel calls “QuickPath Interconnect”, which is equivalent to AMDs Hyper Transport with a few differences. Intel had pretty much reached the upper limits of what it could do with its FSB and with FSB frequencies approaching 1,600 MHz, even faster memory could not alleviate the problem of keeping the CPU fed with data from memory. In fact, it is the IMC that resulted in the changed cache sizes on Nehalem, and not the other way around. Industry experts have rightly criticised Intel for hiding the huge memory latencies associated with having an FSB-based interconnect between CPU and RAM by adopting the brute force approach — increasing the processor cache to sometimes crazy levels. Obviously this is an inefficient, not to mention costly approach. Now, thanks to the IMC, Intel can make Nehalem shine with a relatively small L2 cache. Another trick is the inclusion of a new secondary Translation Look-Aside Buffer (TLB); earlier CPUs made do with a single TLB. The TLB is a CPU cache that is used by memory management hardware (like the IMC) to improve the speed of virtual address translation. Intel also copies AMD’s way of designating the specific processor regions and the Nehalem is divided into two parts termed as “core” and the rather awkward “un-core”. The core includes the physical CPU cores and L1 and L2 caches while the un-core consists of everything else on the chip, including the L3 cache and IMC.

The branch prediction mechanism of the Core i7 has also seen a significant change. A second-level branch predictor with a much larger data set and a deeper history has been added to the prediction unit in all Core 2 processor cores. The deeper history in conjunction with the larger data set allows for broader

searching. Although this secondary predictor unit is slower than the primary unit, its ability to correct incorrect predictions made at the first-level and thus prevent any core overtime should more than compensate for its reduced speed of operation. Then there is a buffer called the “Renamed Returned Stack Buffer” which ensures that correct data is pulled from the processor’s stack, even in the event of an incorrect prediction. Therefore Nehalem beefs up the chances of a hit, and minimises overhead and latency in case of a miss — a huge plus, to be sure. HT together with this two-stage branch prediction should bring immense benefits when working with multiple threads. The branch prediction could also do wonders for single-threaded applications: Since each core’s resources are shared according to the workload, if two threads are pushed at a core then each of them gets one-half of the resources. If a single thread is to be executed, it gets the full attention on the core it is pushed to, giving it greater performance. While on the topic of parallelism, Nehalem can execute more in-flight micro-ops than Penryn — the count is up from 96 to 128. Micro-ops, short for micro operations, are detailed low-level instructions issued to implement complex machine instructions. So the i7 has an increase in parallelism at the instruction level as well; a serious benefit when you consider each core supports HT.

The IMC

Now a little bit on that shiny new IMC. Intel’s IMC supports three channels instead of two. So be prepared to see i7 based motherboards with six memory slots. This IMC supports only DDR3 and up to 24 GB of RAM. This is good for server configurations or workstations needing more than 8 GB of RAM. Although the IMC doesn’t support the fastest DDR3 memory, it doesn’t need to. Firstly, the theoretical bandwidth that tri-channel memory configurations offer, even at memory speeds of 1,333 MHz, is higher than a dual-channel configuration running memory clocked at 1,800 MHz. Besides greater bandwidth than all previous Intel systems, the IMC on each i7 CPU ensures that latencies will be minimal. In fact, slower memory also means tighter timings — which we’ve seen as more conducive to performance than higher memory clock speeds. All this will sound very familiar to most people following processor technology over the past five years — this is exactly what AMD told us about its memory controller — theoretical bandwidth doesn’t matter as much when using an IMC because most of the increased bandwidth gets wasted with FSB latencies. It seems Intel was listening to AMD as well. Thanks to the IMC, multi-CPU systems based on Nehalem processors will consist of a dedicated RAM pool per CPU. If

one CPU needs access to some data which is in another’s data pool the QPI link between both the CPUs will come into play. This may result in some latency, but it should not be a lot. Obviously this won’t affect desktop users with a single CPU.

One of the main talking points of the Nehalem architecture is the QPI link itself. The QPI is a point-to-point interface, so there’s no conflict of interests for bandwidth. The QPI is the means for cores to talk to each other. It also connects the CPU to the motherboard chipset and allows the CPU to talk with PCI Express, USB, and SATA components. So far all the Nehalem desktop CPUs that Intel has released have a single QPI link. Server CPUs will have two links, with the second link allowing crosstalk between CPUs in multi-socket systems. The high-end Nehalem, the i965 Extreme Edition, has a link that allows 6.4 GT per second, (GT is an acronym for giga-transfer, yet another new term!). The lower Nehalems have a QPI link that allows 4.8 GT per second. The total bandwidth offered by each QPI is 25.6 gigabits per second and each consists of two 20-bit, bi-directional links. Note that the total bandwidth of 25.6 gigabits per second is double the bandwidth avail-

Nehalem CPUs			
Model Numbers ->	i965 Extreme Edition	i940	i920
Fabrication Process	45 nm	45 nm	45 nm
Core Speed	3.2 GHz	2.93 GHz	2.66 GHz
Socket	LGA 1366	LGA 1366	LGA 1366
L2 Cache Size	256 KB x 4	256 KB x 4	256 KB x 4
L3 Cache Size	8 MB	8 MB	8 MB
QPI Speed (giga-transfers per sec)	6.4	4.8	4.8
Effective Multiplier	24	22	20
TDP	130 watts	130 watts	130 watts

able on the 1,600 MHz FSB of Intel’s X48 chipset. In a truly modular fashion, Intel can add more QPI’s to a Nehalem CPU if the need arises.

The clock speed of all Nehalem CPUs is a derivative of the base QPI clock which is 133 MHz. This is much lower than the FSB speeds of previous CPUs from Intel; where the base FSB clock was 333 MHz, or even 400 MHz. Therefore all Nehalem CPUs have much higher multipliers. The L3 cache and the IMC operate on a difference clock called the un-core clock. This figure is 20X the base QPI clock i.e. 2.66 GHz. This is similar to AMD’s Phenom design where multiple clocks are used for different parts. In terms of power-management Nehalem differs as well: While Phenom allows each core to request different clock speeds, incidentally the reason why one needs to install the AMD Cool’n’Quiet driver; the Nehalem attempts to run all cores at the same clock speed at all times. The only exception being when Turbo Mode (see below) is enabled. In other words, if a particular core